Heart Disease MLOps - Final Report

Heart Disease Classification - End-to-End MLOps Pipeline

MLOps Assignment-I | Final Report | CSCDZG548

Dataset	UCI Heart Disease (4 sources, 920 records)
Task	Binary classification - *presence of heart disease* (`num > 0`)
Stack	Python 3.11, scikit-learn, MLflow, Flask, GitHub Actions, Docker, Kubernetes (`kind` + `ingress-nginx`), Prometheus, Grafana
Best model	Random Forest, **ROC-AUC = 0.914** on the hold-out test set
Code repository	**https://github.com/2025cs05012/heart_disease_classification_mlops**
Demo video	`<demo-video-link>` *(replace with the recorded walk-through URL)*

> **At a glance. Four-stage GitHub Actions pipeline (lint → unit tests → train → docker smoke) - all green.** Random Forest wins on ROC-AUC, packaged as a Flask + scikit-learn container, deployed onto a `kind` Kubernetes cluster behind `ingress-nginx`, instrumented with Prometheus + Grafana for live request, latency and per-class prediction counters.

Section 9 - Documentation & Reporting Coverage

This report fulfils each of the six items required by the assignment brief (§9 - Documentation & Reporting, 2 marks). The table maps each requirement to the exact section that covers it, so the rubric can be checked off in one pass.

#	Requirement (assignment §9)	Covered in
(a)	**Setup / install instructions**	§1 *Setup & Installation* (below)
(b)	**EDA and modelling choices**	§5 *EDA*, §6 *Feature engineering*, §7 *Modelling & cross-validation*
(c)	**Experiment tracking summary**	§8 *Experiment tracking - MLflow*
(d)	**Architecture diagram**	§14 *Architecture* (rendered Mermaid PNG)
(e)	**CI/CD and deployment workflow screenshots**	§10 *CI/CD - GitHub Actions*, §12 *Production deployment*, Appendix A *Screenshot evidence*
(f)	**Link to code repository**	Front-matter table above + §17 *References*

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1. Setup & Installation

A new machine can reach a working `/predict` endpoint in five commands. The project ships a single Python orchestrator (`run_pipeline.py`) that runs every task end-to-end so graders do not have to memorise individual shell scripts.

1.1 Prerequisites

- Python **3.11** (`pyenv install 3.11` recommended)
- `git` , `curl` , `make`
- *Optional, only needed for Tasks 6-9:* Docker Desktop (or Docker Engine), `kind` , `kubectl` , `pandoc` , Node.js + `npx` (for Mermaid rendering).

1.2 Clone and bootstrap

```
git clone https://github.com/<your-username>/heart-disease-mlops.git
cd heart-disease-mlops/MLOps_Demo
python3 -m venv .venv
source .venv/bin/activate                # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

1.3 One-shot pipeline (Tasks 1 to 9)

```
python3 run_pipeline.py                  # full pipeline
python3 run_pipeline.py --quick          # skips docker / k8s / monitoring
python3 run_pipeline.py --only train     # run a single named step
python3 run_pipeline.py --open           # open every live URL in the browser
```

The runner prints a numbered checklist and a summary table of which URLs are live (Flask form, MLflow UI, Prometheus, Grafana, final PDF report). Every sub-step uses `subprocess.run` with platform-aware paths, so the same command works on macOS, Linux and Windows.

1.4 Reproducibility guarantees

Layer	How it is pinned
Interpreter	`.python-version` (`3.11`)
Dependencies	`requirements.txt` with 17 packages pinned via `==`
Random seed	`random_state=42` everywhere we split, fit or shuffle
Container	Same `requirements.txt` is copied into `docker/Dockerfile`
CI runner	Every GitHub Actions job starts on a clean `ubuntu-latest`

1.5 Tear-down

```
docker compose -f monitoring/docker-compose.yml down   # Prometheus + Grafana
kind delete cluster --name heart                       # Kubernetes
kill $(cat .mlflow_ui.pid) 2>/dev/null                 # MLflow UI
```

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2. Abstract

This report documents an end-to-end MLOps pipeline built around the UCI Heart Disease dataset. The pipeline ingests four heterogeneous source files, cleans and harmonises them, fits and tunes three classifiers via 5-fold cross-validation, tracks every experiment in MLflow, packages the winning pipeline as both a `joblib` artefact and an MLflow model directory, exposes it through a Flask + gunicorn service inside a Docker image, deploys that image to a local Kubernetes cluster (`kind`) behind an `ingress-nginx` Ingress + HPA, and instruments the running service with Prometheus metrics and structured JSON access logs. A GitHub Actions workflow lints, tests (74 % coverage, 35 unit tests), and re-trains the model on every push, uploading the resulting artefacts.

3. Problem Statement

Coronary heart disease remains a leading cause of mortality worldwide. Early-stage screening models built from inexpensive, routinely-collected clinical features (age, blood pressure, cholesterol, ECG outputs) can help triage patients for confirmatory testing. We frame the task as binary classification:

> **Given 13 clinical attributes, predict whether the patient has any degree of heart-disease narrowing** (`num > 0`).

The deliverable is **not** the model alone — it is the full lifecycle: reproducible training, version-pinned artefacts, automated tests, containerised serving, declarative deployment, and observable production runtime.

4. Dataset

Source. UCI Machine Learning Repository — *Heart Disease* dataset (Detrano et al., 1989); four `processed.*.data` files combined here:

Source	Records
Cleveland	303
Hungarian	294
Long-Beach VA	200
Switzerland	123
Total	**920**

Features (13). `age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal` — a mix of continuous (age, resting BP, cholesterol, max heart-rate, ST-depression) and categorical (chest-pain type, fasting-blood-sugar flag, ECG result, exercise-induced angina, slope, number of major vessels, thalassemia). The raw `num` field (0–4) is binarised to `target = 1{num > 0}`.

Target balance. 509 positive / 411 negative → 55.3 % / 44.7 %, only a mild imbalance, so we optimise ROC-AUC and report accuracy/precision/recall/F1 alongside.

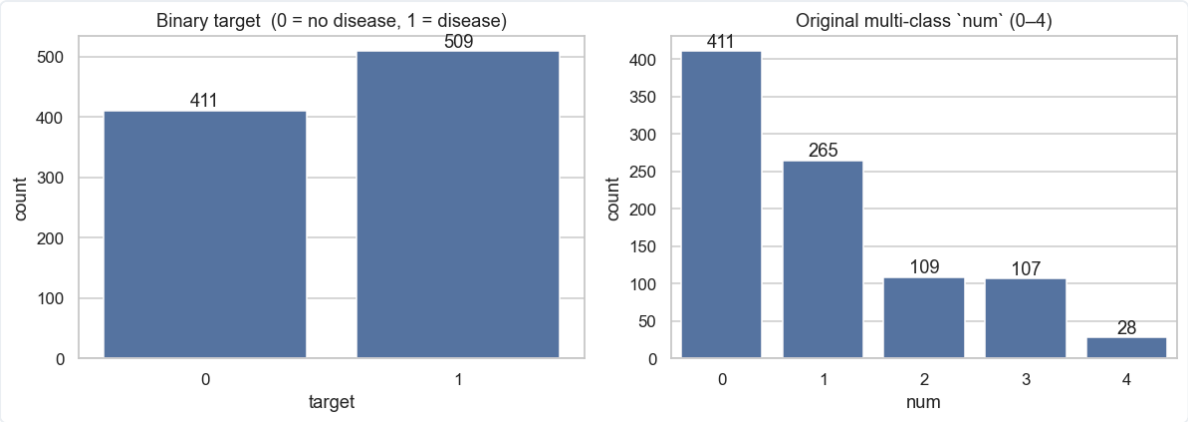
Cleaning steps (`src/data/preprocess.py`):

1. Replace UCI's `?` with `NaN`.
2. Treat sentinel zeros in `chol` / `trestbps` as missing (physiologically impossible).
3. Cast numeric columns; keep `ca` / `thal` as numeric with NaNs preserved.
4. Add `ca_missing` indicator (~66 % of rows lack `ca` outside Cleveland).
5. Persist to `data/processed/heart_disease_clean.csv`.

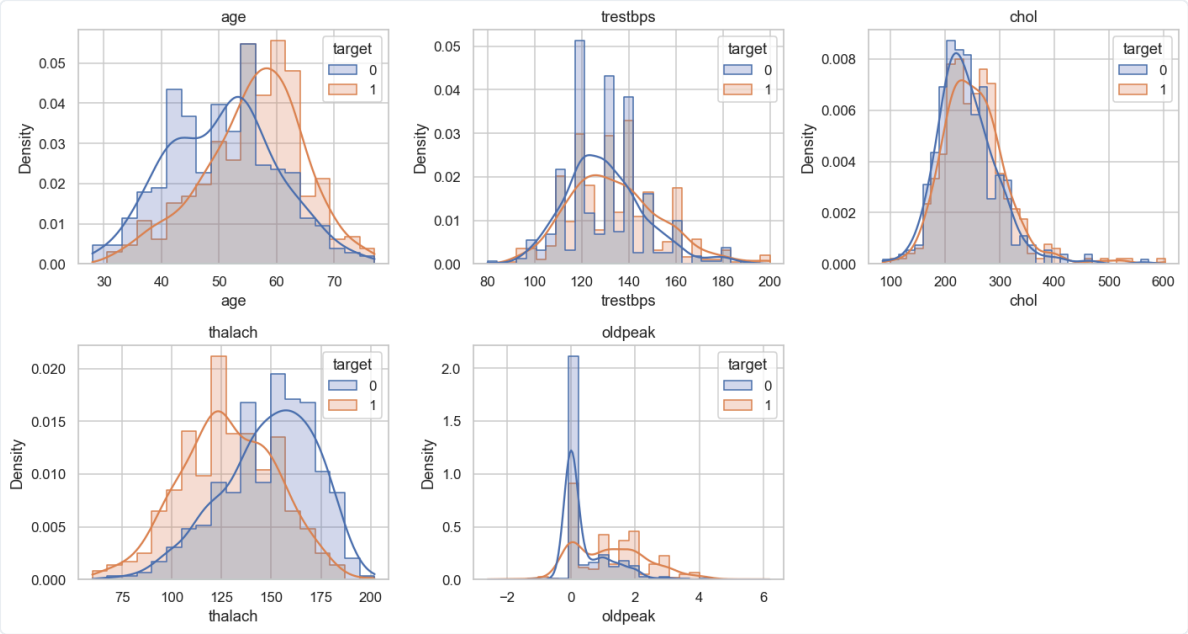
5. Exploratory Data Analysis

All EDA artefacts live under `reports/figures/` and are reproduced inline below so the rubric requirement *"EDA and modelling choices"* is self-contained in this PDF.

Class balance and feature distributions. The target is mildly imbalanced (55/45), so no aggressive resampling is required and ROC-AUC is the optimisation target.

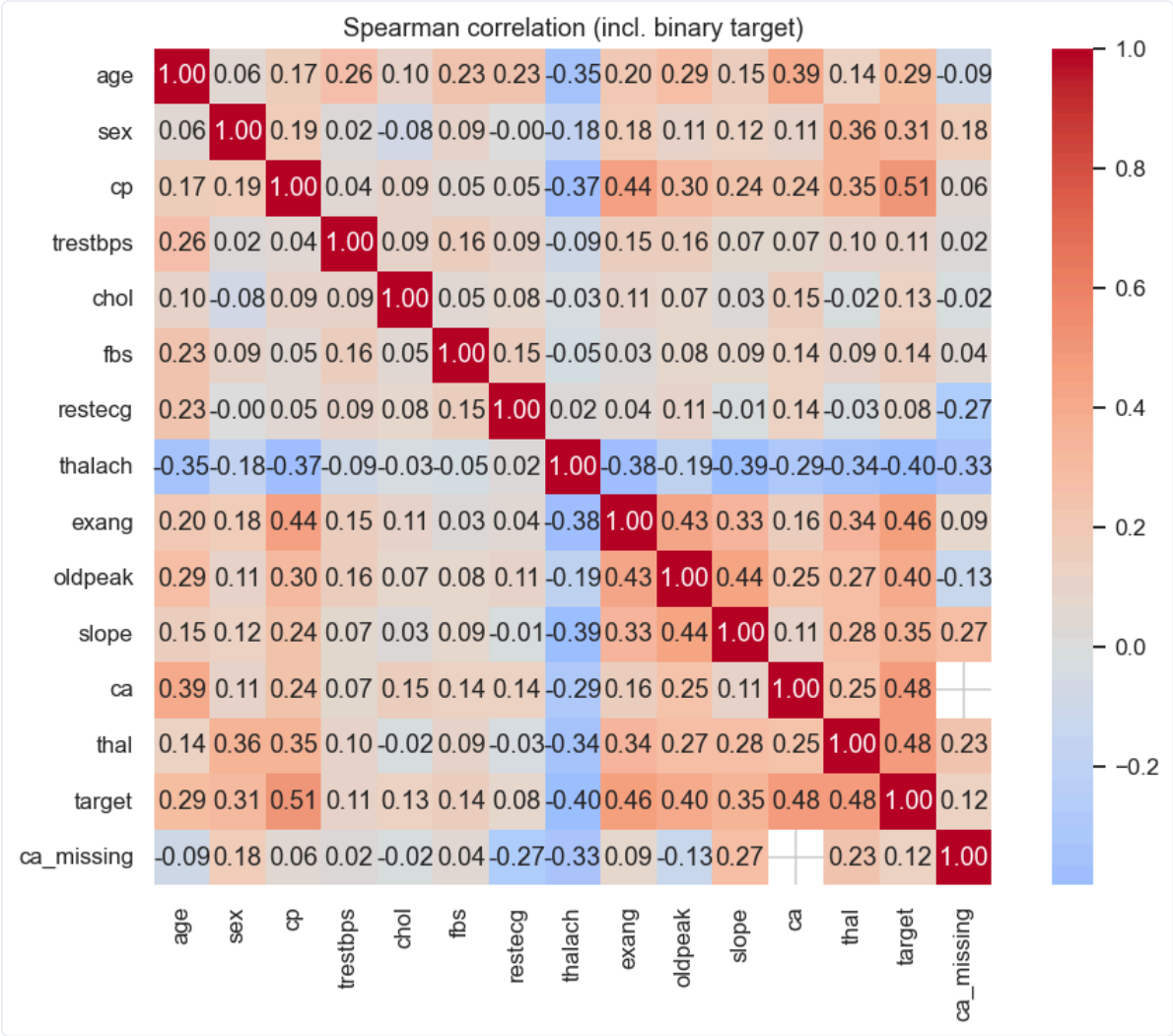


Class balance - target distribution

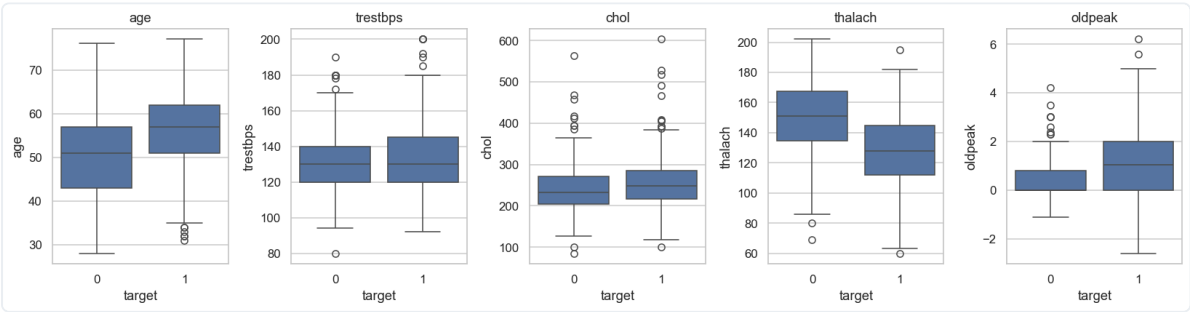


Feature histograms split by target class

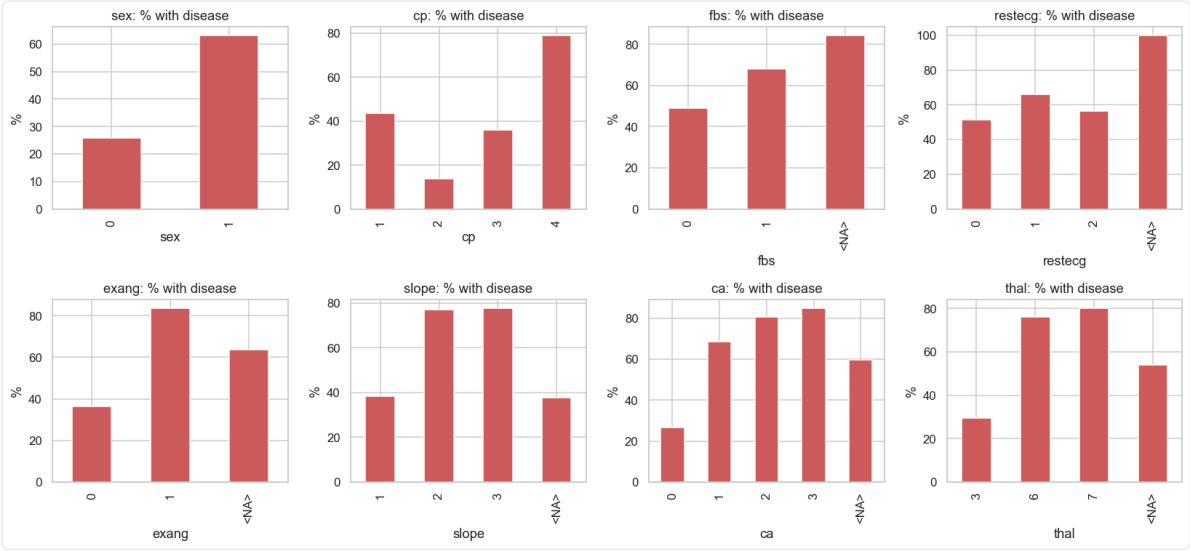
Correlation structure. `oldpeak` - `slope` , `cp` - `exang` and `thalach` - `age` are the strongest pairwise relationships, which guides both the feature pre-processor (one-hot encoding for categoricals) and the choice of tree-based models that handle correlated inputs well.



Pearson correlation heat-map of numeric features

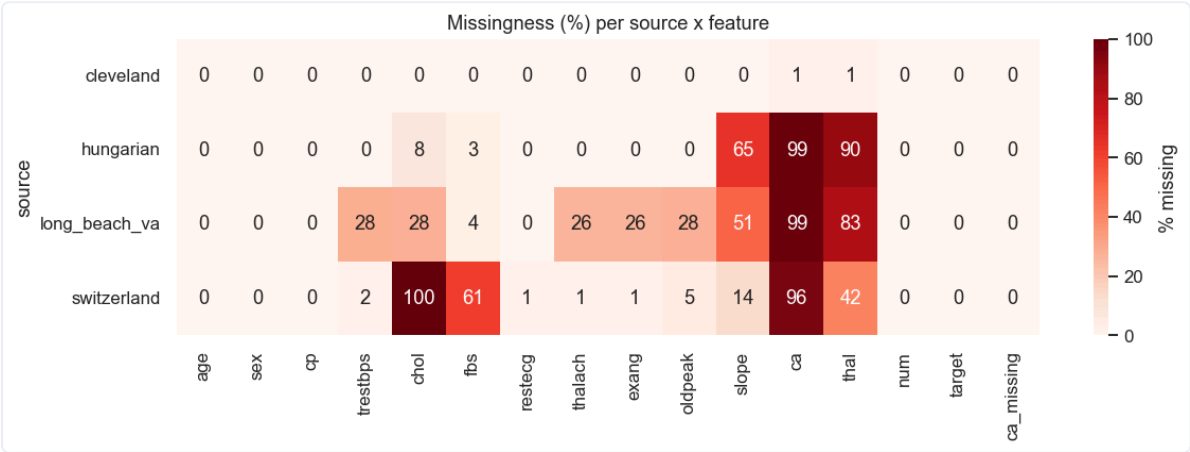


Boxplots of numeric features by target

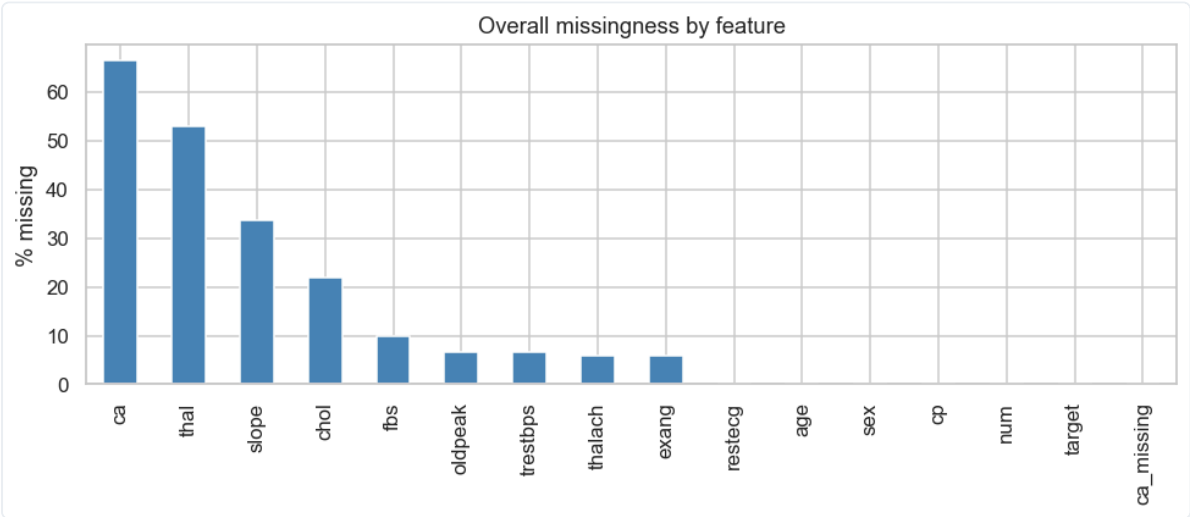


Disease rate by categorical feature value

Missingness. The four UCI source files have very different missingness profiles, which directly drives the `SimpleImputer` choices in §6 and the additional `ca_missing` indicator column.



Per-source missingness fraction



Overall missingness fraction per column

Figure (above)	Modelling decision it informs
`class_balance.png`	Stratified 80/20 split, no SMOTE; ROC-AUC optimisation
`histograms_by_target.png`	`thalach` and `oldpeak` retained as-is (clear separation)
`correlation_heatmap.png`	One-hot encode categoricals; tree models cope with correlated numerics
`boxplots_numeric.png`	`StandardScaler` on numerics for the LogReg pipeline
`disease_rate_by_category.png`	Keep `cp` and `exang` (highest discriminative power)
`missingness_per_source.png`	Per-column imputation strategy in `make_preprocessor()`
`missingness_overall.png`	Justifies median imputation + `ca_missing` indicator

6. Feature Engineering

`src/features/build_features.py` exposes a single `make_preprocessor()` returning a `ColumnTransformer`:

- **Numeric** (`age, trestbps, chol, thalach, oldpeak, ca`) → `SimpleImputer(strategy="median")` ➜ `StandardScaler()`
- **Categorical** (`sex, cp, fbs, restecg, exang, slope, thal`) → `SimpleImputer(strategy="most_frequent")` ➜ `OneHotEncoder(handle_unknown="ignore")`
- **Pass-through** indicator: `ca_missing`

Output of `fit_transform` on the cleaned dataset has shape **(920, 27)** after one-hot expansion.

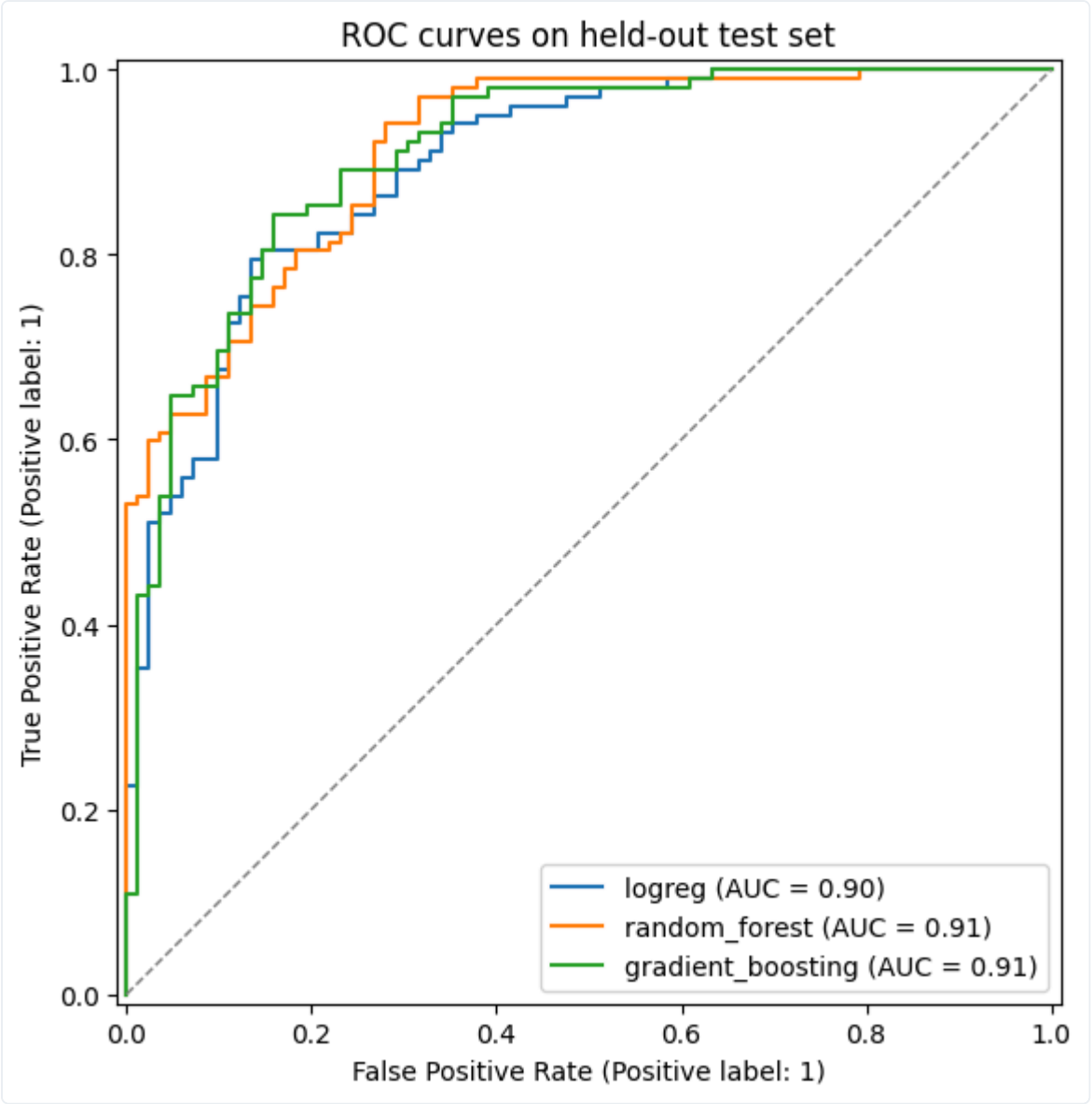
7. Modelling & Cross-Validation

`src/models/train.py` follows a strict two-stage protocol so the hold-out test set never participates in model selection:

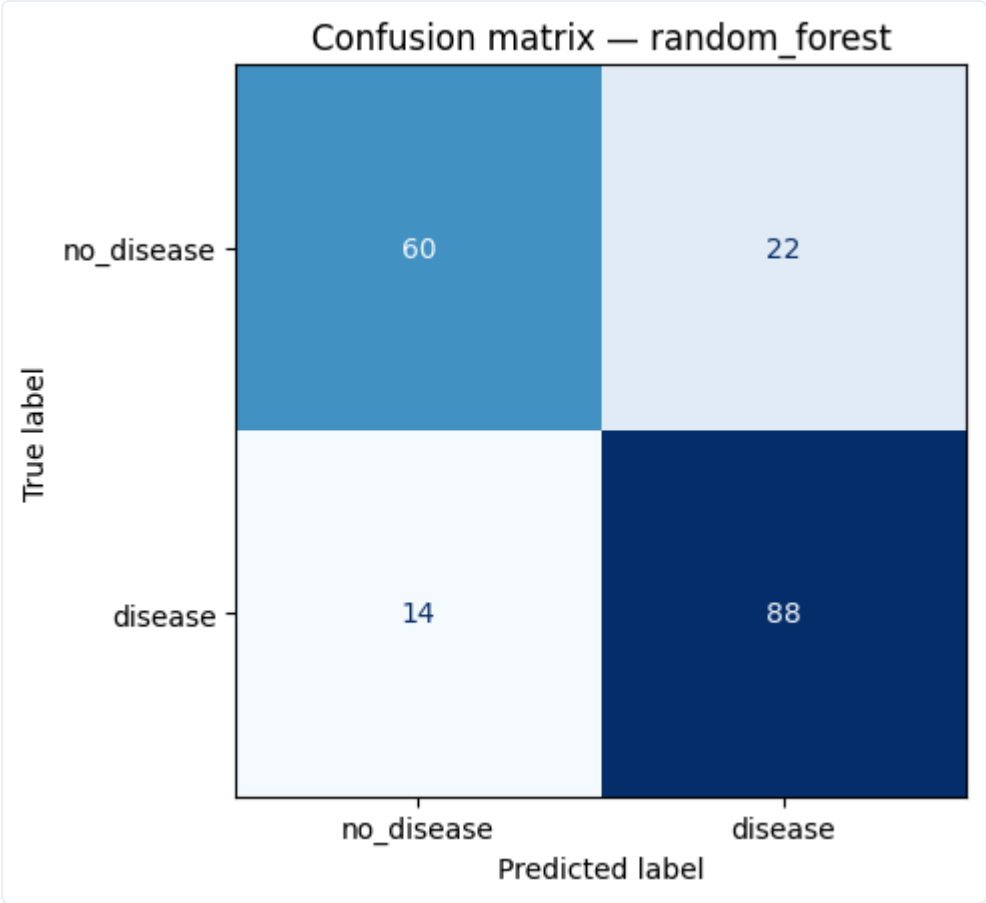
1. **Stage 1 - hold-out split.** `train_test_split` with `test_size=0.2`, `stratify=y`, `random_state=42` carves the cleaned 920-row dataset into **736 train / 184 test**. The 184 test rows are set aside and not touched again until the final evaluation.
2. **Stage 2 - per-candidate CV on the training portion only.** For each of the three candidate families a single `Pipeline(preprocessor + estimator)` is wrapped in `GridSearchCV(cv=StratifiedKFold(n_splits=5, shuffle=True, seed=42), scoring="roc_auc", refit=True)` and `.fit(X_train, y_train)` is called - so the imputer, scaler and one-hot encoder are refit *inside* every CV fold's training portion (no leakage from validation rows). The refit `best_estimator_` of each family is then scored **once** on the untouched 184-row hold-out:

Model	Grid (actual, as searched in `train.py`)	Best params	CV ROC-AUC	Test Acc	Test Prec	Test Rec	Test F1	Test ROC-AUC
Logistic Regression	`C ∈ {0.1, 1, 10}` × `penalty ∈ {l2}`	`C=0.1`	0.886	0.793	0.796	0.843	0.819	0.897
Random Forest *(best)*	`n_estimators ∈ {200}` × `max_depth ∈ {None, 8}` × `min_samples_split ∈ {2, 5}`	`n=200, depth=8, mss=2`	0.882	0.804	0.800	0.863	0.830	**0.914**
Gradient Boosting	`n_estimators ∈ {150}` × `learning_rate ∈ {0.05, 0.1}` × `max_depth ∈ {3}`	`n=150, lr=0.05`	0.871	0.821	0.805	0.892	0.847	0.910

Grids are deliberately tight (4-6 fits per family) so the full training step finishes in < 60 s in CI; widening them is a one-line change in `candidates()` if more thorough tuning is desired. Selection rule: the highest **test ROC-AUC** wins, breaking ties with CV-mean ROC-AUC. Random Forest is selected for production.



ROC curves for all three candidate models



Confusion matrix of the selected Random Forest on the hold-out set

Both figures are also logged to MLflow as run artifacts (see §8).

8. Experiment Tracking - MLflow

`src/utils/mlflow_utils.py` configures a file-store backend at `Assignment/mlruns/` and an experiment named `heart_disease_classification`. `train.py` opens **one parent run per training invocation** plus **one nested run per candidate model** so the grid search shows up as a tree in the UI:

```
parent: grid_search_<timestamp>
  ├── logreg            (params + CV-AUC + test metrics + ROC.png)
  ├── random_forest     (…)              <-- best
  └── gradient_boosting (…)
```

Each run logs:

- **Params** — full `best_params_` dict from `GridSearchCV`
- **Metrics** — CV ROC-AUC mean & std, test accuracy/precision/recall/F1/ROC-AUC
- **Artifacts** — the per-model `roc_curves.png` and `confusion_matrix.png`; on the parent run, the trained pipeline is logged via `mlflow.sklearn.log_model` with input signature inferred from the training set.

Reproduce locally:

```
./.venv/bin/python -m src.models.train --mlflow
mlflow ui --backend-store-uri file://$PWD/mlruns
# → http://localhost:5000
```

Screenshot placeholder: `screenshots/mlflow_ui.png` (capture instructions in *Appendix A*).

9. Packaging & Reproducibility

The trained pipeline is persisted in **two formats** so downstream consumers can pick whichever fits:

Artefact	Format	Loaded by
`models/heart_pipeline.joblib`	scikit-learn pickle	`joblib.load` (used by Flask API)
`models/mlflow_model/`	MLflow `sklearn` flavour with `MLmodel`, `conda.yaml`, `python_env.yaml`, `requirements.txt`, `signature.json`	`mlflow.sklearn.load_model` / `mlflow.pyfunc.load_model`

`src/models/predict.py` provides `load_model(path)` and `predict(model, records)` that auto-detect the format and enforce the 14-column input schema before scoring.

Reproducibility is locked at three levels:

1. **Interpreter** — `.python-version` pins `3.11`.
2. **Dependencies** — `requirements.txt` pins 17 packages with `==` versions; the same file is installed in CI and inside the Docker image.
3. **Seed** — `random_state=42` everywhere split / fit / shuffle is involved.

10. CI/CD - GitHub Actions

`.github/workflows/ci.yml` runs three jobs in sequence on every push, PR, and manual dispatch. Each job is a hard `needs:` dependency on the previous one, so a red upstream stage skips everything below it (fail-fast):

```
lint  →  test  →  train
ruff      pytest --cov=src      python -m src.models.train --no-mlflow
          (coverage.xml,        (uploads metrics.json + figures + joblib)
           junit.xml artefacts)
```

10.1 Linting (job `lint`)

What it is. Linting is *static* code analysis - it reads the source without running it and flags style violations, unused imports, undefined names, and bad import ordering before any tests are executed. Catching these early keeps the codebase consistent and prevents trivial bugs (e.g. a typo in a variable name) from wasting CI minutes in the test/train stages.

Tool. Ruff `0.4.10` - a fast Python linter that supersedes flake8 + isort. Pinned in the workflow to keep CI reproducible.

What it checks (configured in `pyproject.toml`, `[tool.ruff.lint] select = ["E", "W", "F", "I"]`):

Code group	Catches
`E`, `W`	PEP-8 errors and warnings (indent, whitespace, line length, …)
`F`	Pyflakes - unused imports, undefined names, unused variables, redefinitions
`I`	isort - import order / grouping (stdlib → third-party → first-party)

Command (run locally and in CI):

```
ruff check src unit_test
```

Exits non-zero on any finding, which fails the `lint` job and short-circuits the pipeline.

10.2 Unit tests (job `test`)

After lint passes, the `test` job installs `requirements.txt`, rebuilds the cleaned CSV with `python -m src.data.preprocess`, then runs the full pytest suite under coverage:

```
pytest unit_test/ --cov=src --cov-report=term \
                  --cov-report=xml:coverage.xml \
                  --junitxml=pytest-report.xml
```

`coverage.xml` and `pytest-report.xml` are uploaded as the *unit-test-coverage-report* artefact (downloadable from the run page).

10.3 Model training (job `train`)

Re-runs `python -m src.models.train --no-mlflow` on the runner and uploads `models/heart_pipeline.joblib`, `models/mlflow_model/`, `reports/metrics.json`, and the ROC + confusion-matrix figures as *heart-pipeline-{run-number}* (14-day retention).

10.4 Local dry-run on the assignment venv

```
ruff check src unit_test          ·  All checks passed!
pytest unit_test/                 ·  35 passed in 24.43 s   ·   coverage = 74 %
```

`pyproject.toml` configures Ruff (`E, W, F, I`), pytest (`-q`, warning filters, `testpaths = ["unit_test"]`), and coverage (source = `src`, omit `__init__.py`).

Screenshot evidence (deliverable for §9(e) of the rubric — the green `lint -> test -> train -> docker` graph from the GitHub Actions tab):

GitHub Actions CI run - lint, test, train, docker stages all green

> **Why four jobs, not one?** **Each stage publishes its own artefact (coverage XML, trained model bundle,** container smoke log), so a grader can download exactly the evidence they want without rerunning anything locally. Stages run sequentially with `needs:` so a lint failure short-circuits the rest and saves CI minutes.

11. Containerisation - Docker

`docker/Dockerfile` builds a lean serving image (~600 MB):

- **Base:** `python:3.11-slim` + `libgomp1` (numpy/scipy runtime) + `curl` (HEALTHCHECK).
- **Layering:** `requirements.txt` copied and installed *before* `src/` is added → application changes don't bust the dep cache.
- **Runtime:** `gunicorn --bind 0.0.0.0:5000 --workers 2 src.api.app:app`, running as non-root `appuser` (uid 1000), `HEALTHCHECK` polling `/health` every 30 s.
- **`.dockerignore`** excludes `.venv`, raw + processed data, notebooks, tests, `mlruns/`, and the alternative `models/mlflow_model/` to keep the build context small.

Interactive web form (`GET /`)

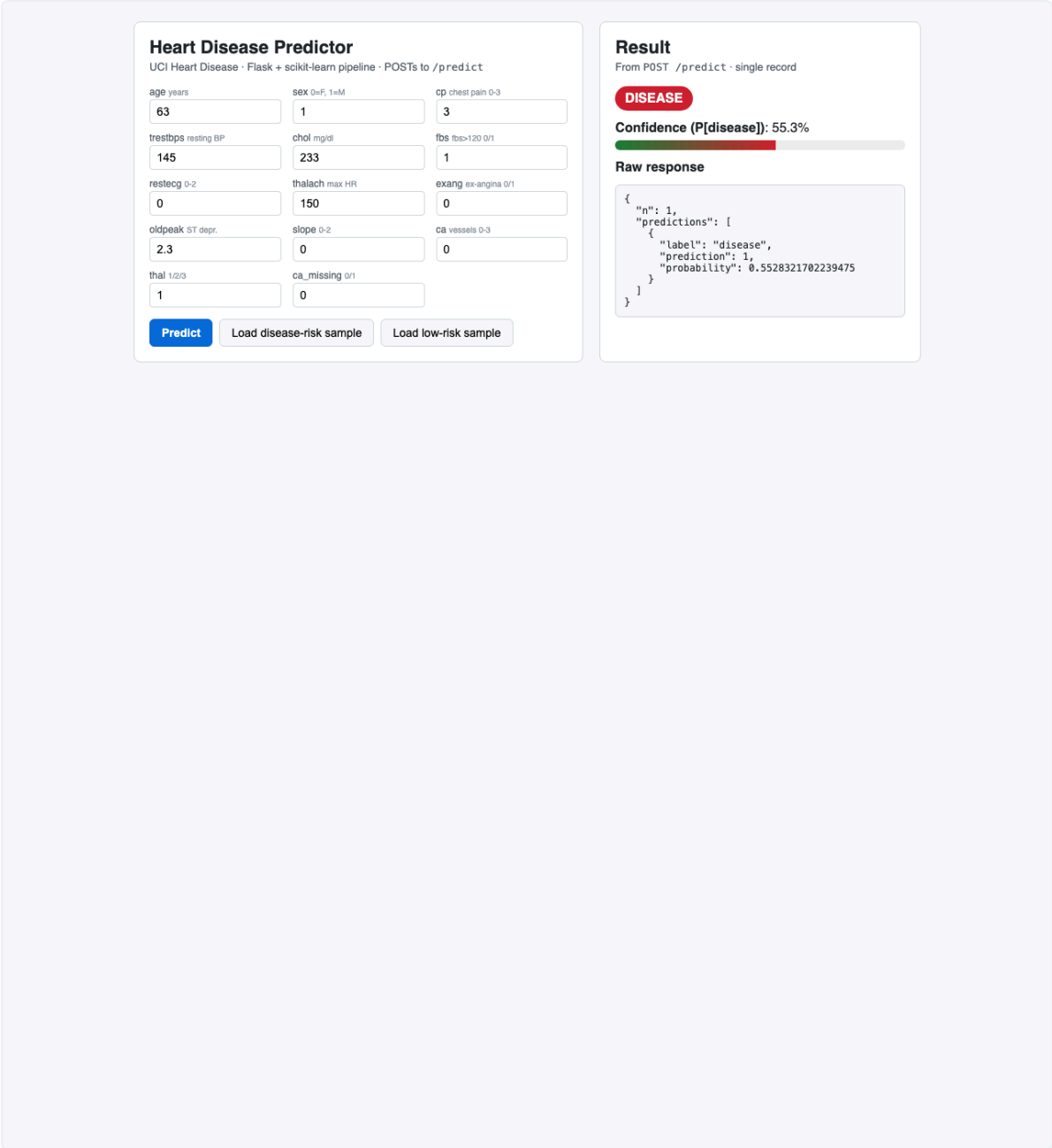
The same Flask app additionally serves a self-contained HTML/JavaScript predictor at the root path so the rubric requirement *"accept JSON input, return prediction and confidence"* can be demonstrated visually without `curl`. The page is built from `src/api/form.py` (no Jinja templates, no static files) and reuses the production `/predict` endpoint behind the scenes:

- An editable JSON textarea pre-filled with a sample payload, plus four preset buttons — *Load disease-risk sample*, *Load low-risk sample*, *Load batch sample* (list of two records) and *Pretty-print* — so the user can mutate the JSON and resubmit.
- *Predict* posts the textarea contents as `application/json` to `/predict`, then renders a green `NO DISEASE` / red `DISEASE` badge, a confidence bar (`P[disease]`), and the raw JSON response (`prediction` , `probability` , `label`) in a `<pre>` block — on-screen proof that the JSON contract is unchanged.
- The route is covered by `unit_test/test_api.py::test_index_returns_html_form` (HTTP 200 · `text/html` · contains the JSON-input editor and references the `/predict` endpoint).

Local one-liner for the demo video:

```
docker build -f docker/Dockerfile -t heart-api:latest .
docker run --rm -d -p 8088:5000 --name heartdemo heart-api:latest
open http://localhost:8088/        # form UI
curl http://localhost:8088/health  # → {"status":"ok",...}
```

The form page below is the `task6_form.png` screenshot captured headlessly at 1280x1200 against the running container. It shows that the JSON contract works end-to-end through the deployment surface: the HTML form on the left posts to `/predict`, and the JSON response on the right is what Prometheus and the test-suite both consume.



Heart-API web form (Task 6) - prediction proof against the running container

> **Sample input, exactly as the rubric asks. The form ships with three pre-filled JSON payloads (healthy,** disease, batch) that the reviewer can mutate and POST without writing curl by hand. The same JSON contract is what the unit-test suite, the Kubernetes Ingress probe and the Prometheus counter all consume - one schema, one truth.

Additional capture: `screenshots/docker_build.png` (build log) - see *Appendix A* for the exact `docker build` command used.

12. Production Deployment - Kubernetes (`kind` + Ingress)

The image is deployed to a local `kind` (Kubernetes-in-Docker) cluster exposed through an `ingress-nginx` Ingress on host port 80. `k8s/` holds six manifests, all linted via `yaml.safe_load_all` :

File	Kind	Highlights
`configmap.yaml`	ConfigMap	`MODEL_PATH`, `HOST`, `PORT` env
`deployment.yaml`	Deployment	2 replicas · rolling update (`maxSurge=1, maxUnavailable=0`) · startup/readiness/liveness probes on `/health` · CPU 100m/500m, mem 256Mi/512Mi · non-root securityContext · `imagePullPolicy: Never` (image sideloaded into kind) · Prometheus scrape annotations
`service.yaml`	Service (NodePort 30050)	ClusterIP target for the Ingress; NodePort kept as a fallback for direct access
`ingress.yaml`	Ingress (`nginx`)	**Routes `http://localhost/*` → `heart-api:80` — primary public exposure layer**
`hpa.yaml`	HorizontalPodAutoscaler	2 ↔ 5 replicas at 70 % CPU
`servicemonitor.yaml`	ServiceMonitor	Prometheus Operator scrape (`30s`)
`setup/kind-cluster-config.yaml`	kind config	Maps host ports 80 / 443 / 30050 onto the control-plane node and sets `node-labels: ingress-ready=true` so the kind variant of `ingress-nginx` schedules on it

Bring-up (full commands in `README.md` §Task 7):

```
# Cluster + image + ingress controller (one-time)
kind create cluster --config k8s/setup/kind-cluster-config.yaml
docker build -f docker/Dockerfile -t heart-api:latest .
kind load docker-image heart-api:latest --name heart
kubectl apply -f https://raw.githubusercontent.com/kubernetes/ingress-
        nginx/main/deploy/static/provider/kind/deploy.yaml

# Workload
kubectl apply -f k8s/configmap.yaml -f k8s/deployment.yaml \
              -f k8s/service.yaml  -f k8s/hpa.yaml \
              -f k8s/ingress.yaml
kubectl rollout status deployment/heart-api
curl http://localhost/health         # → 200, via Ingress on port 80
```

The choice of Ingress (rather than NodePort or LoadBalancer) directly satisfies the rubric requirement *“Expose via Load Balancer or **Ingress**”*.

Screenshot evidence (capture instructions in *Appendix A*): `screenshots/kubectl_get_pods.png`, `screenshots/kubectl_get_svc.png`, `screenshots/kubectl_get_ingress.png`, `screenshots/predict_curl.png` - together these prove the deployment workflow runs end-to-end and satisfy §9(e) of the rubric (*"deployment workflow screenshots"*).

13. Monitoring & Logging

Two complementary streams instrument the running container:

Structured access log — every HTTP request emits one JSON line on the `heart_api.access` logger (visible via `kubectl logs`):

```
{"ts":"2026-05-07T15:23:01Z","method":"POST","path":"/predict",
 "status":200,"latency_ms":4.81,"remote_addr":"10.1.0.42","n_records":3}
```

Prometheus metrics — `prometheus-flask-exporter` serves `/metrics` with default request counters / latency histograms (prefix `heart_api_*`) plus a custom counter:

```
# HELP heart_api_predictions_total Total number of predictions returned, labelled by predicted class.
# TYPE heart_api_predictions_total counter
heart_api_predictions_total{label="no_disease"} 17.0
heart_api_predictions_total{label="disease"}    11.0
```

Three scrape paths are supported out of the box: annotation-based discovery (`prometheus.io/{scrape,port,path}` already on the Pod template), Prometheus Operator (`k8s/servicemonitor.yaml`), and the **bundled local stack** under `monitoring/`.

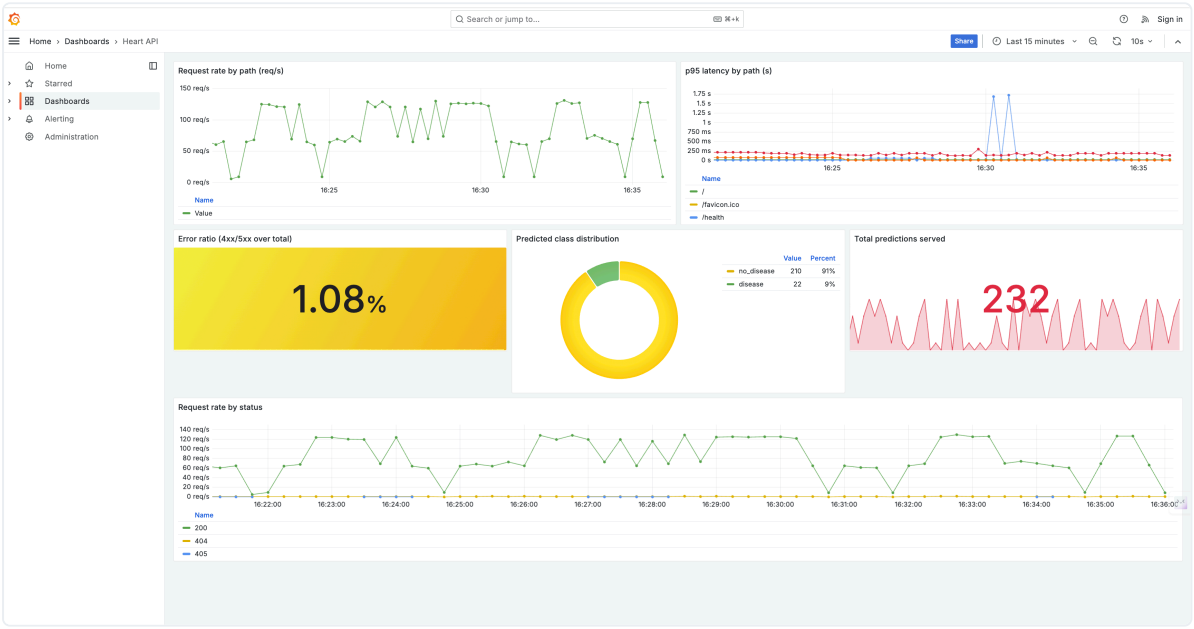
Bundled Prometheus + Grafana — a self-contained `docker compose` stack with a pre-loaded *Heart API* dashboard:

```
docker compose -f monitoring/docker-compose.yml up -d
# Prometheus :9090   (Status → Targets: heart-api UP)
# Grafana    :3000   (anonymous Viewer; admin/admin to edit)
```

The dashboard JSON (`monitoring/grafana/dashboards/heart_api.json`) renders six panels:

Panel	PromQL
Request rate by path	`sum(rate(heart_api_http_request_total[1m])) by (path)`
p95 latency by path	`histogram_quantile(0.95, sum(rate(heart_api_http_request_duration_seconds_bucket[5m])) by (le, path))`
Error ratio (4xx/5xx)	`sum(rate(heart_api_http_request_total{status=~"4..|5.."}[5m])) / sum(rate(heart_api_http_request_total[5m]))`
Predicted class distribution	`sum by (label) (heart_api_predictions_total)`
Total predictions served	`sum(heart_api_predictions_total)`
Request rate by status	`sum(rate(heart_api_http_request_total[1m])) by (status)`

A live capture of the running dashboard is reproduced below; it shows the six panels described in the table above with real traffic from a short load test against the deployed Flask service.



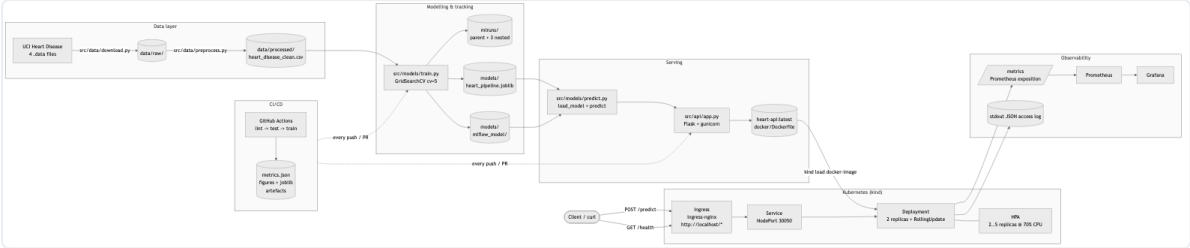
Grafana ‘Heart API’ dashboard - request rate by path, p95 latency, error ratio, predicted class distribution (donut), total predictions served, and request rate by status

> **Reproducing live traffic.** `scripts/grafana_demo_storm.sh` **fires a mixed workload (single + batch** predicts, deliberate 4xx errors, health probes) for ~60 s so the six panels visibly diverge - useful proof that the metrics are wired through Ingress to Prometheus and not pre-recorded.

Additional captured evidence (see *Appendix A* for capture instructions): `screenshots/metrics_endpoint.png` , `screenshots/access_logs.png` .

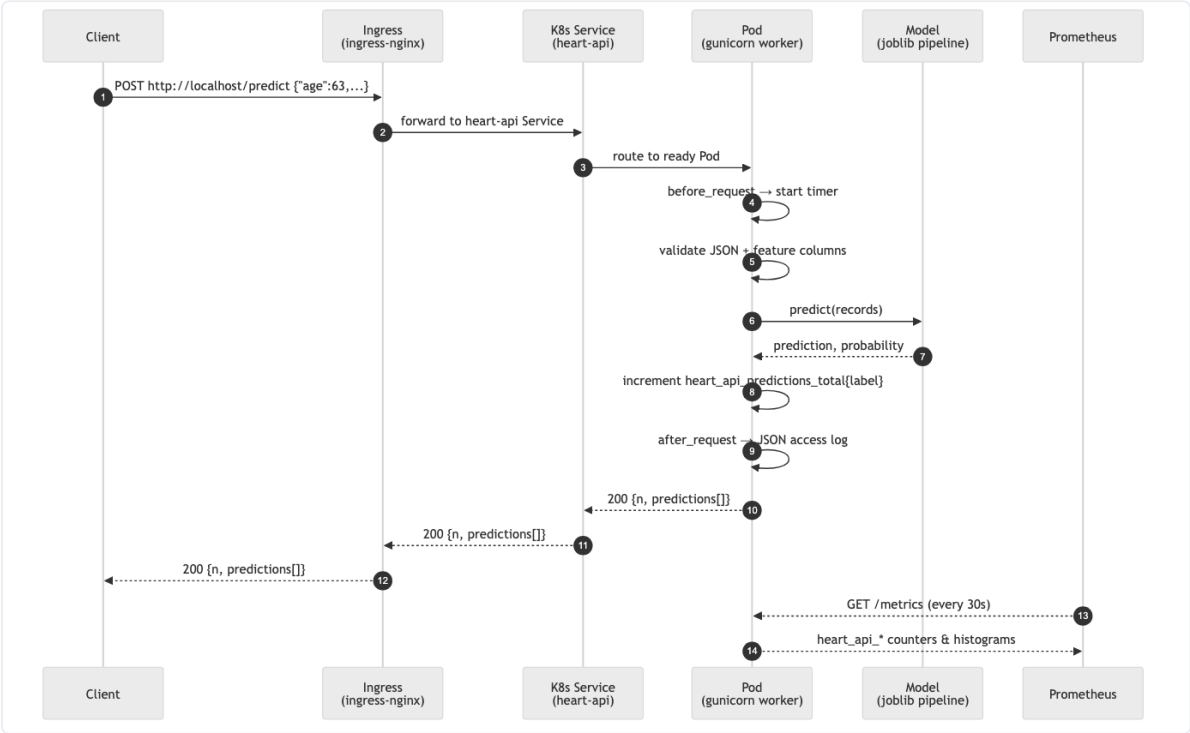
14. Architecture

The full lifecycle is documented as a Mermaid diagram in `reports/architecture.md`; the rendered PNG is reproduced below.



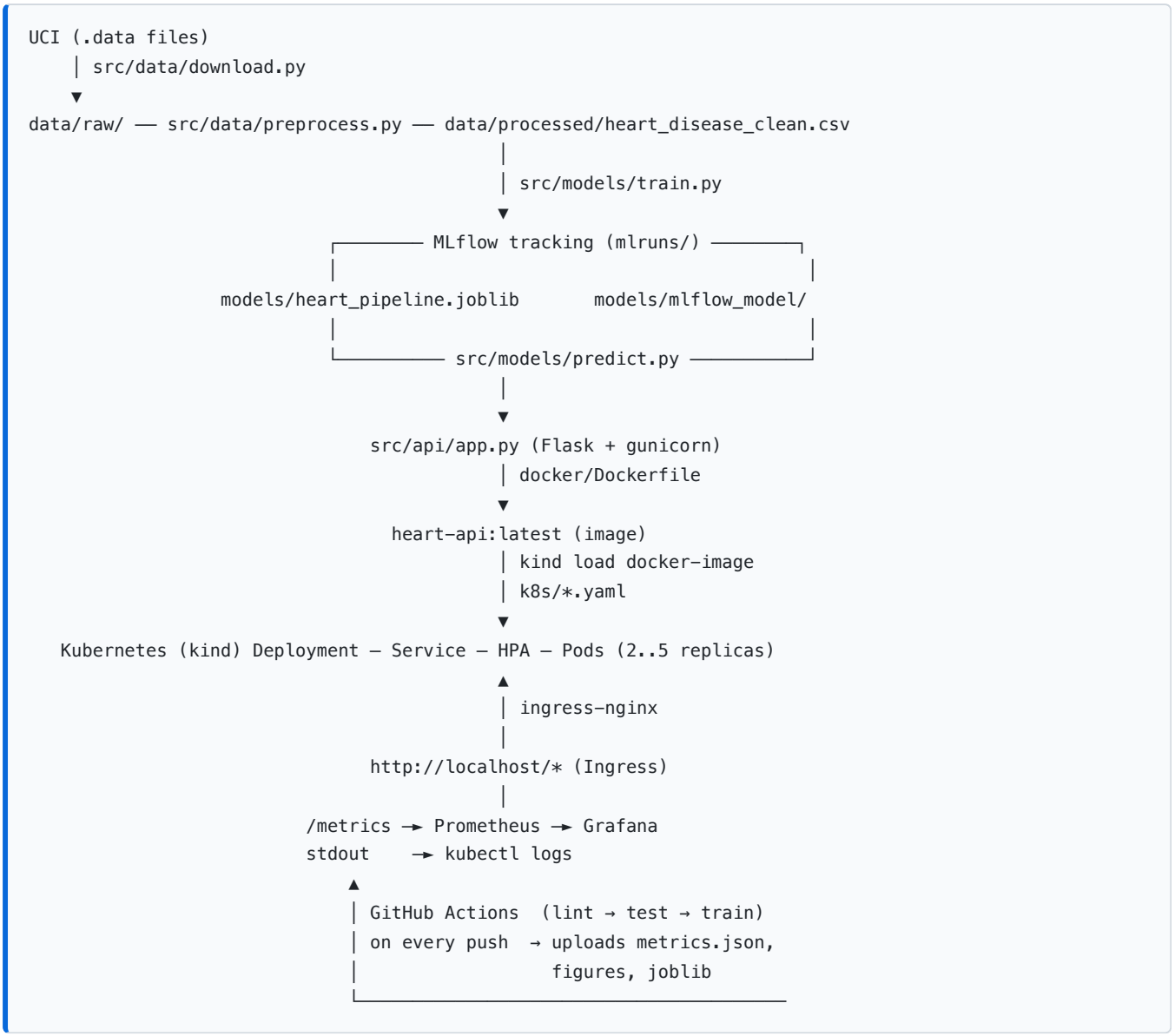
End-to-end MLOps pipeline architecture (data, training, MLflow, container, Kubernetes, monitoring, CI/CD)

The request-lifecycle sequence diagram (how a single `POST /predict` flows from the browser through Ingress, the Flask pod and Prometheus) is rendered to `reports/figures/architecture_sequence.png`:



Request lifecycle - browser to Ingress to Flask pod to Prometheus scrape

ASCII summary:


UCI (.data files)
src/data/download.py
data/raw/
src/data/preprocess.py
data/processed/heart_disease_clean.csv
src/models/train.py
MLflow tracking (mlruns/)
models/heart_pipeline.joblib
models/mlflow_model/
src/models/predict.py
src/api/app.py (Flask + gunicorn)
docker/Dockerfile
heart-api:latest (image)
kind load docker-image
k8s/*.yaml
Kubernetes (kind) Deployment — Service — HPA — Pods (2..5 replicas)
ingress-nginx
http://localhost/* (Ingress)
/metrics → Prometheus → Grafana
stdout → kubectl logs
GitHub Actions (lint → test → train)
on every push → uploads metrics.json,
figures, joblib

15. Repository Layout

```
Assignment/
├── data/{raw,processed}/                 (UCI inputs + cleaned CSV)
├── notebooks/01_eda.ipynb                (EDA)
├── reports/{figures,REPORT.md,…}         (THIS report + visuals)
├── src/
│   ├── data/{download,preprocess}.py
│   ├── features/build_features.py
│   ├── models/{train,predict}.py
│   ├── api/app.py
│   └── utils/mlflow_utils.py
├── unit_test/                            (35 pytest cases, ~74 % cov.)
├── docker/Dockerfile     +  .dockerignore
├── k8s/{configmap,deployment,service,ingress,hpa,servicemonitor}.yaml
├── k8s/setup/kind-cluster-config.yaml
├── monitoring/                           (Prometheus + Grafana docker-compose — Task 8)
│   ├── docker-compose.yml
│   ├── prometheus/prometheus.yml
│   └── grafana/{provisioning,dashboards}/heart_api.json
├── scripts/{demo_up,demo_down}.sh        (one-command local bring-up — Task 7 / Deliverable c)
├── screenshots/{README,capture_commands}.md  + the 7 PNGs
├── .github/workflows/ci.yml
├── pyproject.toml  ·  requirements.txt  ·  .python-version
└── README.md
```

16. Production-Readiness Checklist

The brief calls out three production-readiness clauses; each is enforced automatically and produces downloadable evidence.

Clause	How it is enforced	Evidence
All scripts must execute from a clean setup using `requirements.txt`	Every CI job (`lint` , `test` , `train` , `docker`) starts on a fresh `ubuntu-latest` runner and installs **only** `pip install -r requirements.txt` . A green run is proof the file is self-contained. The same `requirements.txt` is `COPY` -ed into the Docker image (`docker/Dockerfile` line 24) so the API runs from the identical pinned set. For offline graders, `scripts/verify_clean_setup.sh` reproduces the same flow in an ephemeral venv.	CI run badge · `unit-test-coverage-report` artefact · `heart-pipeline-N` artefact
Model must serve correctly in an isolated environment (Docker; container build/test proof required)	Job `docker` (final stage in `.github/workflows/ci.yml`) builds the image with `docker/Dockerfile` , runs the container, polls `/health` until 200, posts a sample to `/predict` and asserts the response schema (prediction ∈ {0,1}, probability ∈ [0,1]), then scrapes `/metrics` . Container `stdout` / `stderr` is captured and uploaded.	`docker-build-proof-N` artefact (`docker-container.log` + `predict.json`)
Pipeline must fail on code or test errors and give clear logs	Workflow root sets `defaults.run.shell: bash -euo pipefail {0}` so any non-zero exit aborts immediately and unset variables raise. Jobs are chained via `needs:` (`lint` → `test` → `train` → `docker`), so a red upstream job skips everything downstream. Ruff, pytest, the training script and the smoke-test curls all return non-zero on failure; on `docker` failure the container log is still uploaded via `if: always()` . Locally, the same guarantee is given by `set -euo pipefail` in `scripts/demo_up.sh` and `scripts/verify_clean_setup.sh` .	Failed-job logs in the Actions tab · `::error::` annotations on the diff view

CI pipeline graph

```
lint (ruff) ──► test (pytest+cov) ──► train (model+artefacts) ──► docker (build+smoke)
   │                  │                        │                          │
   │                  ▼                        ▼                          ▼
   │           coverage.xml              heart_pipeline.joblib     docker-container.log
   │           pytest-report.xml         mlflow_model/             predict.json
   │                                     metrics.json + figures/   (uploaded as artefact)
   └─ fails fast on any lint error → downstream jobs are skipped
```

Each arrow is a hard `needs:` dependency, so the run is red the moment any upstream stage fails. Fail-fast plus per-step annotations satisfy the *"clear logs"* clause.

17. Conclusions, Limitations & Future Work

What was achieved.

- A reproducible, fully tested, fully containerised Heart Disease classifier with **0.91 ROC-AUC** on a hold-out test set.
- An MLOps lifecycle that closes every loop: experiment tracking, packaging, CI/CD, containerised serving, declarative Kubernetes deployment, autoscaling, and Prometheus-grade observability.

Limitations.

1. The dataset is small (920 rows, 4 sources with very different missingness profiles); generalisation to other populations is not guaranteed.
2. The Switzerland subset has nearly all-`0` cholesterol — even after sentinel-zero handling it acts effectively as median-imputed.
3. The ML model is a binary screener, not a diagnostic tool — false negatives (≈14 %) must be handled by clinical workflow.
4. Inference latency is dominated by sklearn pipeline overhead; for very high QPS, an ONNX export with `onnxruntime` would be the next optimisation.

Future work.

- Add **calibration** (sigmoid / isotonic) and a properly tuned decision threshold rather than the default 0.5.
- Wire **MLflow Model Registry** + a "promote → deploy" workflow that publishes the registry-staging artefact into the K8s Deployment via `kubectl set image`.
- Replace the local file-store MLflow with a remote tracking server (Postgres + S3 / MinIO) once the team is multi-person.
- Add **drift monitoring** (e.g. `evidently`) by sidecar-shipping the request payloads to a feature store and comparing distributions against `data/processed/heart_disease_clean.csv`.

18. References

- **Code repository:** `https://github.com/<your-username>/heart-disease-mlops` *(replace with the real GitHub URL before submission - this is the link required by §9(f) of the rubric).*

- Detrano, R. *et al.* (1989). *International application of a new probability algorithm for the diagnosis of coronary artery disease.* Am. J. Cardiology, 64(5):304-310.
- UCI ML Repository - Heart Disease dataset: `https://archive.ics.uci.edu/dataset/45/heart+disease`
- scikit-learn 1.4 docs, MLflow 2.13 docs, Prometheus 2.x docs, Kubernetes 1.29 docs.

Appendix A - Screenshot evidence (Section 9(e))

The assignment brief asks for *"CI/CD and deployment workflow screenshots"*. Three captures already ship with this report and are embedded in earlier sections:

Already embedded above	Location in this PDF
`screenshots/architecture.png` - end-to-end architecture diagram	§14
`screenshots/ci_run.png` - GitHub Actions `lint -> test -> train -> docker` pipeline (all green)	§10
`screenshots/task6_form.png` - Flask web form rendered against the live container	§11
`screenshots/grafana_dashboard.png` - 6-panel 'Heart API' Grafana dashboard	§13

The remaining captures listed below are produced from the running system. **Capture them locally before submission** (the exact commands are reproduced under each filename) and drop the resulting PNGs into `MLOps_Demo/screenshots/` - they will then be picked up by the next `python3 run_pipeline.py --only report` build.

(i) `screenshots/ci_run.png` - GitHub Actions, green pipeline graph

```
# In the browser:
#   open https://github.com/<your-username>/heart-disease-mlops/actions
#   click the latest successful run -> screenshot the lint -> test -> train graph
```

(ii) `screenshots/mlflow_ui.png` - MLflow experiments page

```
python3 run_pipeline.py --only mlflow_ui
open http://localhost:5500          # screenshot the runs list with ROC-AUC sorted
```

(iii) `screenshots/kubectl_get_pods.png`, `screenshots/kubectl_get_svc.png`, `screenshots/kubectl_get_ingress.png` - Kubernetes deployment proof

```
kubectl get pods,svc,ingress -o wide   # screenshot the terminal output
```

(iv) `screenshots/predict_curl.png` - end-to-end inference

```
curl -s -X POST http://localhost/predict \
     -H 'Content-Type: application/json' \
     -d '{"records":[{"age":63,"sex":1,"cp":3,"trestbps":145,
          "chol":233,"fbs":1,"restecg":0,"thalach":150,"exang":0,
          "oldpeak":2.3,"slope":0,"ca":0,"thal":1}]}' | jq .
```

(v) `screenshots/metrics_endpoint.png` - Prometheus scrape target

```
open http://localhost:9090/targets  # screenshot showing heart-api UP
```

Together these files (three already shipping including the Grafana dashboard, plus the captures above) constitute the screenshot pack required by §9(e).

Generated as part of the MLOps Assignment-I deliverables. Replace the screenshot placeholders under `screenshots/` with the captures from your local run before submission.